Structural Learning Mechanics: A Preregistered Protocol for Learner-Conditioned Structural Data Allocation

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Abstract

Curriculum, active, influence-based, saturation-aware and model-aware data selection are now well-developed research areas: existing systems already condition example selection on a learner's evolving state, target missing skills, deduplicate semantically redundant data, and reallocate training budget after apparent saturation. This paper does *not* claim to have invented any of those functions, and it does not report that a proposed extension of them works. Instead it isolates a narrower, distinct residual hypothesis — that data allocation conditioned on an explicit, *directional*, quantity-of-interest- and applicability-boundary-scoped *structural* coordinate, with a learner-specific *vector* mastery state and exact reuse of the same structural identities at inference, allocates training budget more cost-effectively than a strong static structural curation and than the strongest generic learner-conditioned parent, subject to noncompensatory retention and transfer-safety constraints — and it freezes, in advance, the experimental protocol designed to answer whether that residual exists. We state the conceptual mechanics that are ready: three separate projections of one substrate (scientific authority, search routing, training allocation) that must not share authority semantics; a learner-conditioned, staleable structural-saturation receipt; and a mastery vector over principle, composition, boundary, representation, transfer and retention coordinates. We argue several *architectural* propositions — learner-conditioned saturation is not a fixed point of the training operator, a scalar mastery summary is representationally inadequate for the allocation decision, retention safety must be a hard constraint rather than an objective term, and the training projection cannot mint scientific authority. We then preregister a Phase 0–4 protocol, mapping each phase to its governing decision record, with frozen design parameters, decision gates, and an explicit falsifier for each. The paper makes zero empirical claims. Whether the structural residual survives measurement is an open question the protocol is built to answer, and it may return negative; the protocol is designed so that a negative result absorbs the mechanism back into the sibling paper on method evolution rather than manufacturing a spurious contribution.

Status: design-and-protocol preprint; no empirical validation has been performed. This document is a *preregistered design and protocol*, not an empirical paper. No training run reported here has been executed: there is no accuracy, effect size, cost measurement, or capability outcome in this paper, and none is claimed. Every quantitative value that appears is a *preregistered threshold, target, or design parameter* chosen *before* any data access, and is labelled as such. The empirical claims sketched here are conditional on (i) authorization of a training programme through decision records #461, #466, #467 and #468, (ii) availability of a trainable, sufficiently capable in-ladder model and authorized high-performance compute, and (iii) an affirmative authorization at decision gate #462. Until those conditions are met, the residual hypothesis this protocol targets is *unresolved*, and a negative outcome is an explicitly supported result of running it.

1 The gap: what is prior art, and what is the residual

A learner that is trained on a corpus does not benefit equally from every example. This observation is old, and the modern literature that acts on it is large and strong. Skill-based frameworks learn ordered skill dependencies and sample adaptively along them [1]; skill-graph selection curates mathematical pretraining data by structural role [8]; model-aware influence methods maintain an evolving estimate of which data most improve the current parameters [7]; token-level and reference-loss methods select *within* an example [5]; influence-for-instruction-tuning methods pick data by optimizer-aware gradient alignment [6]; document de-duplication and diversification methods show that intelligent repetition can beat random one-pass exposure [3]; semantic de-duplication removes near-duplicate content at web scale [4]; domain-reweighting methods optimize the training mixture [2]; and skill-targeted adaptive training profiles a learner's *missing* skills and synthesizes or selects examples against them after vanilla fine-tuning saturates [9]. Compositional-generalization work shows that atomic-skill examples do not automatically teach composition and that the distribution shape and training format matter [10]. Work on the emergence and test-time use of abstract structure in language models shows that structural information is learned and used at inference [11], and reusable-experience methods route and compose latent skills across adaptation [12]. The cognitive-science literature, finally, establishes that spaced and varied retrieval improves *retention and transfer*, not merely acquisition [13].

We concede all of this as prior art. In particular, this paper does *not* claim: the first learner-conditioned curriculum; the first missing-skill targeting; the first saturation-aware adaptive training; the first model-state-specific allocation; or that "the model's current state changes which data are useful." Each of those is occupied. Any paper that framed its contribution as one of them would be correctly rejected as a re-description of [1, 7, 9].

The residual. What is *not* obviously occupied is a specific conjunction. ORION already represents transfer at inference through directional, quantity-of-interest- and applicability-boundary-scoped relational structural objects with explicit records of *non-preserved* properties and hostile near-misses (the sibling method-evolution and structural-transfer mechanics; §6). The residual hypothesis of this paper is that data allocation conditioned on *that same* structural substrate — rather than on a free skill label, a token loss, an influence scalar, or a domain group — carries distinct, measurable value. Concretely, the residual is the joint presence of:

1. explicit *directional* / quantity-of-interest / applicability-boundary structural coordinates as the unit of allocation, not a free skill label;
2. a learner-specific *vector* saturation state distinguishing principle acquisition from composition, boundary, representation and transfer mastery;
3. a *causal* allocation value, established by a matched contrast against a strong static structural curation, not by beating random or full-data training;
4. *exact* reuse of the same structural identities at training-allocation time and at inference-time transfer;
5. an *acceptable total cost*, accounting for structure extraction, mastery probing and scheduling overhead, not tokens alone; and
6. *noncompensatory* retention and transfer safety, so that saved tokens cannot be bought with forgetting or lost composition.

This is an open question, and it may return negative. Whether that residual survives is an *empirical* question, and the honest prior is uncertain. The strongest single threat is skill-targeted adaptive training [9], which already profiles learner-specific missing skills and adapts after saturation; if a faithful instance of it explains the same gain, the residual is not real.

Model-aware influence selection [7] blocks any claim that adaptivity itself is the contribution. De-duplication-and-diversification results [3] and the retention literature [13] forbid modelling saturation as "mastered $\Rightarrow$ never show again." The protocol below is designed precisely so that a negative outcome is reportable and consequential: if adaptive structural allocation does not distinguishably beat *static* structural allocation, or if a simpler parent fully explains any gain, the mechanism is *unsupported as a distinct contribution* and is absorbed into the sibling method-evolution paper rather than spun into a standalone claim.

What this paper is. This paper is therefore a preregistered design-and-protocol preprint. It contributes (a) the conceptual mechanics of learner-conditioned structural data allocation, which are architecturally ready, and (b) the frozen Phase 0–4 experimental protocol that will be executed once training is authorized. It reports no result. It reserves no priority over the prior art above. It states, in advance, the conditions under which its own central hypothesis would be declared false.

2 Structural coordinates and learner-conditioned saturation

2.1 Three projections of one substrate

Let R_t denote the complete external research substrate at time t — a canonical/raw corpus together with scientific and experience state. ORION already exposes a scientific-authority projection $K_t = \pi_{\text{epi}}(R_t)$ (what is scientifically licensed) and a search/routing projection $Q_t = \pi_{\text{search}}(R_t)$ (what should be inspected next). The present mechanics add a third:

$$L_t \;=\; \pi_{\text{train}}(R_t, \theta_t),$$

the *training projection*: which registered example or structural coordinate should receive gradient budget next, given the current model parameters θ_t. The three projections may share content identities and lineage, but they must not share authority semantics. The binding invariant is

$$\text{training utility} \;\neq\; \text{search rank} \;\neq\; \text{scientific authority}.$$

A high-value training example may be scientifically weak; a highly authoritative source may be redundant for a given learner; a centrally cited source may be useful for routing yet add almost no independent structural exposure. The training projection is the only one of the three that depends on θ_t, and that dependence is the source of every architectural property below.

2.2 Structural objects as the unit of allocation

The unit of allocation is not a free-text skill label but a registered relational *structural object s* carrying a direction (which relation licenses which use), a quantity of interest, an applicability boundary, and an explicit set of *non-preserved* properties and hostile near-misses. This is the same object family used for inference-time transfer in the sibling mechanics; the training projection annotates existing structural identities rather than minting new ones, and raw examples remain canonical.

Definition 1 (Structural mastery vector)**.** *For a registered structural object s and parameters θ_t, the learner-conditioned mastery state is the vector*

$$M_t(s) = \big(m_{\text{principle}},\, m_{\text{composition}},\, m_{\text{boundary}},\, m_{\text{representation}},\, m_{\text{transfer}},\, m_{\text{retention}}\big),$$

each coordinate estimated by a frozen probe family *rather than by model introspection: base relational-rule acquisition; use of s in registered multi-structure compositions; behaviour under boundary/regime change and semantic near-misses; invariance across notation and surface encoding; structure-known/domain-new performance; and resistance to forgetting after training moves elsewhere.*

Definition 2 (Learner-conditioned structural redundancy and staleable saturation). *For an example x mapping to structural object(s) $s(x)$, redundancy is defined* relative to the current state, *$R_t(x \mid \theta_t, M_t)$, not as a static pairwise property. A saturation receipt binds a checkpoint/optimizer identity, the structure and coordinate set, a frozen probe-suite hash, exposure/diversity history, the current mastery vector with uncertainty, a marginal-gain estimate, a minimum repetition/retention floor, and an expiry rule; it is invalidated or refreshed when θ_t moves materially, and it grants no scientific authority.*

2.3 Architectural propositions

The following are *architectural* facts about the design — consequences of the definitions and of the product-order discipline inherited from the epistemic mechanics — not empirical findings. They constrain what a correct implementation must look like; they do not assert that the mechanism improves training.

Proposition 1 (Learner-conditioned saturation is not a fixed point of training). *Let the training operator carry $\theta_t \mapsto \theta_{t+1}$ with $\theta_{t+1} \neq \theta_t$. Then in general*

$$\pi_{\text{train}}(R_t, \theta_{t+1}) \neq \pi_{\text{train}}(R_t, \theta_t),$$

so a saturation decision valid at θ_t need not remain valid at θ_{t+1}, even with R_t fixed.

Proof. π_{train} is defined as a function of θ_t through the mastery vector of Definition 1, whose coordinates are outputs of probes evaluated on the current parameters. A non-trivial parameter update changes those outputs on at least one coordinate for at least one structure unless the update is orthogonal to every probe, which is not guaranteed. Hence the projection is time-varying under training. In particular a receipt asserting saturation of s at θ_t carries no guarantee at θ_{t+1}; this is why Definition 2 makes the receipt staleable and checkpoint-bound rather than permanent. The contrast with the scientific-authority projection K_t is exact: a change in L_t driven by a weight update does not, by itself, change any claim in K_t. □

Proposition 2 (A scalar mastery summary is representationally inadequate for allocation). *Let allocation depend on distinguishing a state in which a principle is mastered but its composition is not from a state in which every coordinate is uniformly half-mastered. A scalar summary $\bar{m}(s) \in \mathbb{R}$ cannot separate these states, whereas the vector $M_t(s)$ of Definition 1 can. Hence no scalar mastery score is a sufficient statistic for the allocation decision.*

Proof. Choose two mastery vectors with equal mean, e.g. $(1, 0, \ldots)$ (principle mastered, composition absent) and $(\frac{1}{2}, \frac{1}{2}, \ldots)$. Any function of $\bar{m}$ alone assigns them the same allocation, yet the correct next-example choice differs: the first calls for composition/boundary exposure, the second for continued broad exposure. Therefore the allocation policy is not measurable with respect to $\bar{m}$, and the vector representation is necessary. A scalar is admissible only as a derived reporting summary that never erases a low coordinate, exactly as the product-order discipline requires. This is the training-time image of the epistemic no-scalarization principle. □

Proposition 3 (Retention and transfer safety must be hard constraints, not objective terms). *Let the allocation objective be tokens-to-target for a fixed capability probe, and suppose retention enters as an additive reward term with any finite weight. Then there exists a schedule that improves the objective while driving the retention coordinate arbitrarily low. Therefore retention (and, symmetrically, composition/boundary transfer) safety must be encoded as a noncompensatory constraint — a separate product coordinate with a floor — rather than as a term traded off inside the objective.*

Proof. If retention contributes additively with weight $w < \infty$, then for any target gain Δ on the primary objective there is a reallocation that redirects budget away from the saturated structure, increasing the primary term by more than w times the retention loss it induces; the weighted sum rises while retention falls. Iterating drives retention toward its infimum. A finite weight thus cannot protect the coordinate. Only a hard floor — a constraint that renders any violating schedule infeasible regardless of primary gain — prevents saved tokens from being purchased with forgetting. The de-duplication-with-diversification and spaced-retrieval literatures make this a live, not hypothetical, failure mode [3, 13]. □

Proposition 4 (Training-projection non-interference)**.** *Let ρ be any transition confined to the training projection (annotation, mastery-probe update, saturation-receipt issuance, allocation-policy change), and let π_{epi} be the scientific-authority projection. Then*

$$\pi_{\text{epi}}(\rho(R_t)) = \pi_{\text{epi}}(R_t),$$

unless a separately registered evidence, refutation, specification or review transition also occurs.

Proof. By construction every training surface exposes `grants_scientific_authority == False`, and the scientific-authority projection is defined only over evidence, refutation, specification, novelty and review transitions. A training-only transition touches none of these coordinates, so π_{epi} is invariant under it. Consequently a scheduler that learns a source family yields high transfer gain does not thereby raise the scientific authority of that family's claims; source authority may act only as a minimum data-quality constraint on candidates, never as a multiplier into a training-utility score. This mirrors the routing-only projection invariant of the verified-discovery mechanics. □

3 Adaptive versus static allocation: the causal question

3.1 The estimand

The scientific object is not "does adaptive allocation help?" but a specific contrast. Consider matched arms at equal total compute: **A** uniform/random; **B** semantic diversity/de-duplication [3, 4]; **C** the strongest faithful learner-conditioned parent [9, 7, 1]; **D** *static* ORION structural curation; **E** *adaptive* ORION structural-mastery allocation; with optional **F** a loss/perplexity-only adaptive scheduler and **G** an influence-only scheduler [6].

Proposition 5 (Identification of the residual)**.** *The primary estimand $E - D$ is the unique contrast among these arms that isolates the learner-conditioned structural residual over the static structural baseline. The contrasts $E - A$ and $E - B$ conflate the residual with the value of* any *curation; $E - C$, $E - F$ and $E - G$ identify whether the residual adds anything beyond, respectively, generic learner-conditioning, loss-based selection, and influence-based selection. No single contrast other than $E - D$ answers the paper's residual question, and no positive $E - A$ licenses a residual claim.*

Proof. D and E share the identical structural substrate, cost accounting and constraint set and differ only in whether allocation is conditioned on the learner-specific mastery vector; hence their difference removes every common cause except learner-conditioning and identifies its marginal effect. A and B differ from E in the presence of structural curation itself, so $E - A$ and $E - B$ are confounded by curation value. C, F and G each hold a different rival selection principle fixed, so their contrasts identify the residual beyond that specific rival, not the residual as such. The claim follows. □

3.2 Primary endpoint and typed inference

The primary endpoint is *total cost to a frozen structural-OOD capability target*, reported as a vector: tokens-, examples-, FLOPs-, GPU-hours- and wall-clock-to-target, *plus* all preprocessing, selection and probe cost. A run that trains on fewer tokens but consumes more total compute after structural extraction and probing is not an efficiency win. Secondary endpoints include novel-composition accuracy, boundary/regime robustness, semantic-decoy rejection, representation invariance, cross-domain transfer, calibration, and forgetting/retention. Outcomes are reported as *typed inference states* — `DISTINGUISHABLE_BENEFIT`, `DISTINGUISHABLE_HARM`, `MEASURED_BUT_INDISTINGUISHABLE`, `UNDERPOWERED`, `CANNOT_IDENTIFY`, `INVALID_CONTAMINATED` — never a sign-only verdict. A material minimum detectable effect, a precision target and stop rules are declared before confirmatory access (see §4 for the frozen values, all design parameters).

3.3 Failure-driven, not heuristic-driven, adaptation

A flat or harmful transfer result must not directly spawn a new sampling heuristic. Instead it enters a typed diagnosis chain: `failure` → typed `diagnosis` → bounded, diagnosis-bound policy challenger → fresh assurance → promote/reject/narrow. The diagnosis taxonomy distinguishes at least `SAME_STRUCTURE_SATURATED`, `COMPOSITION_GAP`, `BOUNDARY_GAP`, `REPRESENTATION_GAP`, `TRANSFER_DOMAIN_GAP`, `RETENTION_FORGETTING_GAP`, `MODEL_OPTIMIZATION_FLOOR`, `STRUCTURE_EXTRACTION_DEFECT`, `PROBE_INSTRUMENT_DEFECT`, `DATA_QUALITY_DEFECT` and `INSUFFICIENT_POWER`, because these imply different interventions and because a model floor or an instrument defect must never be mistaken for genuine structural saturation. A counterfactual repair (freeze the failed case; hold model, optimizer, candidate pool and compute fixed; alter exactly one registered mechanism; require the intervention to move the registered failure quantity in development before any challenger is built) blocks the `bad-batch` → `clever-curriculum` → `same-benchmark-retest` → `victory` pathway.

4 The preregistered protocol

The protocol is frozen here, before any training data access. Phases map to their governing decision records; the standalone paper is authorized only if decision gate #462 returns affirmatively after the programme. All numeric values in this section are *design parameters chosen before data*, not measurements.

Frozen design parameters (chosen before outcome). The exposure schedule per structural family is the geometric ladder $n \in \{1, 2, 4, 8, 16, 32, 64, \dots\}$. Mastery is measured on the six coordinates of Definition 1 by probe families frozen before any adaptive-scheduler outcome. The primary estimand is $E - D$ (Proposition 5) on total cost to a frozen structural-OOD target. The registered material effect size is a minimum detectable effect of 0.05 on the paired primary contrast, with interval estimates by item bootstrap at 20,000 resamples; the co-primary false-promotion target — the fraction of arms that reach a positive residual verdict without a genuine effect — is 0. Any broad cross-family or cross-model law requires agreement across at least four exact-verifier structural families for the core comparison and at least six families before a law is asserted. Gold structure labels are assigned by the generator/verifier, never by a language-model judge. Seeds, generators, thresholds, probe-suite hashes and the candidate pool are frozen and their hashes published before the confirmatory seed executes.

4.1 Phase 0 — known-world structural generator (#461)

The programme does not begin with a large model or natural literature. It begins with a generator whose latent ground truth is exact, over independent factors (principle, composition graph, boundary/regime, representation, transfer/domain shell, surface variation), across at least five family types. The generator's own validity is a prerequisite gate: surface wording may not leak the structure label; valid and invalid cases share templates where possible; labels are computed from executable relations rather than perturbation names; coordinate-ablated twins test whether a classifier is merely reading generator artifacts; and task partitions are chronology-bound before any learner outcome. If those checks fail, no downstream learning claim is licensed.

4.2 Phase 1 — structural exposure curves (#461)

For each family, train at the frozen exposure ladder and, at each checkpoint, measure the six mastery coordinates and the marginal value of the next equal-cost example under each of same-structure repetition, new representation, new boundary, new domain shell, new composition, and hostile near-miss. Monotonic decline is *not* required — composition or boundary changes may produce gain spikes, and this is anticipated. The central mechanism is supported only if the *relative* value of equivalent repetition falls after principle mastery while unsaturated coordinates remain valuable. If it does not, the scheduler is not built.

4.3 Phase 2 — powered adaptive versus static (#466)

Only if Phase 1 shows a measurable state-dependent residual is the smallest adaptive `TrainingAllocationPolicy` implemented, with one diagnosis-driven update family, and the matched arms run at equal total compute. The primary estimand is $E - D$; parent-residual estimands are $E - C$, $E - F$, $E - G$; retention, composition and boundary safety are hard constraints, not objective terms (Proposition 3). The decisive preregistered test is that saved tokens must not be bought with a violated constraint: an arm that reaches the cost target while failing a retention or composition floor is a failure, not a win. The negative outcome — `MEASURED_BUT_INDISTINGUISHABLE`, `DISTINGUISHABLE_HARM`, or full explanation by a simpler parent — absorbs the mechanism into the method-evolution paper.

4.4 Phase 3 — exact train/inference structural-identity reuse (#467)

The trained model is frozen and the *exact* registered structural identities used for training allocation are reused in inference-time retrieval and witnessed transfer, over held-out novel compositions, boundary flips, representation changes, domain-new shells and structure-wrong decoys. Strong unification requires one structural-identity family to serve both training allocation and inference transfer. If a separate latent skill representation is required at inference, the honest report is two mechanisms, not a unification — a result that weakens, but does not by itself refute, the residual.

4.5 Phase 4 — cross-family / cross-model generalization law (#468)

The powered comparison is repeated across at least six structural families and across model scales and families, with a sign test across families and ΔC reported so that blocking regressions dominate. Only agreement across families and scales licenses a generalization law; a residual that appears in one setting and vanishes or reverses in others is reported as scoped, never as a law. Throughout, system-level uplift (parallel probing, exhaustive rival enumeration, externalized memory) is attributed as a system win via a ΔC vector and never reported as "the model learns better."

5 Threats to validity and why the paper may return negative

This protocol is built to be able to fail, and several honest routes to a negative outcome are load-bearing.

The residual may not exist. If Phase 1 finds no state-dependent structural residual, or Phase 2 finds $E - D$ indistinguishable, the learner-conditioned structural mechanism is unsupported. This is a supported result, not a disappointment to be engineered away.

A simpler parent may explain everything. Skill-targeted adaptive training [9], model-aware influence selection [7], loss-based selection [5] or influence-based selection [6] may reproduce any gain. The parent-residual estimands $E - C$, $E - F$, $E - G$ exist precisely to detect this, and a full-explanation outcome terminates the standalone claim.

The saturation model may be wrong. If "mastered $\Rightarrow$ reduce marginal exposure" harms retention or composition despite the floor, the de-duplication-and-diversification and spaced-retrieval evidence [3, 4, 13] is vindicated over the naive picture, and the constraint set — not the objective — was the real contribution.

The substrate may not unify. If Phase 3 requires a separate inference representation [12, 11], the shared-substrate claim is downgraded to two mechanisms.

Cost may erase the token win. Structure extraction, mastery probing and scheduling overhead may exceed the token savings; the total-cost endpoint is defined so this counts as no win.

Instrument circularity. A mastery probe that reads generator artifacts, or an adaptive scheduler tuned on the development set and retested on the same benchmark, would manufacture a spurious residual. Coordinate-ablated twins, chronology-bound partitions, frozen probe hashes, disjoint fresh assurance and the counterfactual-repair contract are the guards; a development instrument must first survive a coordinate-ablated-twin circularity attack, and failed instruments are preserved as negative history.

Power and identification. Underpowered or contaminated runs return `UNDERPOWERED` or `INVALID_CONTAMINATED`, never a sign-only claim. Open-world non-certifiability applies: no finite protocol certifies the residual absolutely; a positive result is bounded relative to the declared families, models and horizon, and any newly observed structural regime reopens it.

6 Relationship to the rest of the series

This paper is a sibling, not a superset, of the surrounding work, and it is careful not to re-claim any of it.

Paper I (epistemic/governance mechanics) contributes the product-order, no-scalarization and freeze-before-outcome discipline that the training projection reuses. Learner-conditioned saturation is deliberately kept *separate* from epistemic saturation: a change in the training projection driven by a weight update alters gradient allocation but, by Proposition 4, changes no scientific claim. Training efficiency is not part of Paper I's contribution.

Paper II (whole-framework architecture / structural transfer) contributes the directional structural objects and witnesses that the training projection annotates. The distinction from external-state operation is exact: there, $\theta_{t+1} = \theta_t$; here the training-time extension permits

$\theta_{t+1} \neq \theta_t$, requiring separate training-state, checkpoint and optimizer provenance. The current framework does *not* already change weights, and this paper does not imply it does.

Paper III (method-evolution mechanics) is the closest sibling and the default owner of this material. Paper III studies experience→method evolution and reports a *negative* two-arm training precursor: a frozen reset-versus-learning packet produced schema-valid records but *zero registered successes* under a small instruction-tuned model, a scoped negative transfer-success result, not a continual-learning gain [17]. That negative precursor belongs to Paper III and is *not* evidence for the present mechanism; it is cited here only to underline that no training gain is on the table. The governance default is that this material remains in Paper III unless the Phase 0–4 residual survives with substantial distinct novelty.

Paper V (verified discovery in mathematics) contributes the routing-only-projection precedent that Proposition 4 mirrors, and the discipline that proposal fluency never becomes authority [17]. Conceptual analogy to avoiding repeated mathematical rediscovery is explicitly *insufficient* to move this paper's claim; only a direct training experiment would.

Paper VI (generative mechanics, prospective) is where value/promise and severity coordinates would decide *whether* a training programme is worth its compute; this paper supplies the allocation object that such a decision would schedule, but makes no claim on that prospective work.

7 What must happen before any empirical claim

No empirical statement in the residual family may be made until all of the following hold.

1. **Authorized compute.** High-performance training capacity must be authorized and executed; the protocol targets an academic HPC environment. No such run has been performed.
2. **A trainable, capable in-ladder model.** A model meeting the registered capability floor must be available and in-scope. The relevant frozen gate flags — `CAPABLE_MODEL_AVAILABLE` and `MODEL_CAPABILITY_FLOOR` — must be satisfied before Phase 2 opens; the sibling capability campaign was previously exhausted without opening the powered comparison, so the confirmatory packet remains *unavailable* rather than promoted from a precursor.
3. **Programme execution in order.** Decision records #461 (Phase 0/1), #466 (Phase 2), #467 (Phase 3) and #468 (Phase 4) must be executed, each gate passing before the next opens, with the Phase-0 generator validity and Phase-1 exposure-curve residual as hard prerequisites.
4. **Authorization at gate #462.** A standalone empirical Paper IV may be produced only if decision gate #462 authorizes it. If #462 returns `ABSORB_INTO_PAPER_III`, `CONCEPTUAL_CROSS_PAPER_ONLY` or `REJECT_NEW_PAPER`, no standalone empirical Paper IV is manufactured, and this document remains a design-and-protocol preprint.

Until every condition is met, the residual hypothesis is unresolved, the mechanics of §2 stand only as architecture, and the protocol of §4 stands only as a frozen design.

Code, materials and AI-use disclosure

A public research artifact accompanies this series. This paper reports no experiment; it preregisters one. Language models and coding agents were used for literature search, drafting and implementation assistance and are not authors, independent reviewers or human experts. No prospective quantity in this paper is reported as a measurement; every numeric value is a preregistered threshold, target or design parameter fixed before data access, and no result is claimed.

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Table 1: Preregistered Phase 0–4 protocol. Every parameter is a design choice fixed before data; no cell reports a measurement.

Phase	**Gate**	**Objective and design**	**Falsifier / negative outcome**
0 — Known-world generator	#461	Build a generator with exact latent ground truth over independent factors (principle, composition, boundary, representation, transfer shell, surface variation); small algebra/rule, graph-flow, coordinate-transform, finite-state and conservation families; labels from executable relations, with coordinate-ablated twins and chronology-bound partitions.	Generator validity fails: surface wording leaks the structure label, or a coordinate-ablated twin classifier matches the full classifier, i.e. the task is readable from artifacts rather than structure. Protocol halts; no learning claim proceeds.
1 — Exposure curves	#461	Train each family at the frozen exposure ladder; at each checkpoint measure all six mastery coordinates and the marginal value of the next equal-cost example when it is same-structure vs. new representation/boundary/domain/composition or a hostile near-miss.	The *relative* value of equivalent repetition does *not* fall after principle mastery while unsaturated coordinates stay valuable, i.e. no state-dependent structural residual exists. Central mechanism unsupported; do not build the scheduler.
2 — Powered Adaptive vs. Static	#466	Matched arms A–E (optional F, G) at equal total compute; primary estimand $E - D$; parent-residual estimands $E - C$, $E - F$, $E - G$; MDE 0.05; typed inference states; retention/composition/boundary encoded as hard constraints.	`MEASURED_BUT_INDISTINGUISHABLE` or `DISTINGUISHABLE_HARM` on $E - D$, or a simpler parent ($C/F/G$) fully explains any gain. Residual unsupported as a distinct contribution; absorb into the method-evolution paper.
3 — Shared train/inference identity reuse	#467	Freeze the trained model; reuse the *exact* registered structural identities/roles/relations/invariants/boundaries in inference-time retrieval and witnessed transfer; hold out novel compositions, boundary flips, representation changes, domain-new shells and structure-wrong decoys.	A separate latent representation is required at inference, i.e. training-allocation and inference-transfer do *not* share one structural identity family. Report two mechanisms, not a unification.
4 — Cross-family / cross-model law	#468	Repeat the powered comparison across ≥ 6 structural families and across model scales/families to test whether the residual generalizes into a stable law rather than a single-setting artifact; sign test across families; report ΔC with blocking regressions dominant.	The residual does not replicate across families or scales, or reverses sign; no generalization law is asserted, and any single-setting effect is reported as scoped, not general.